Towards Post-Lithium Ion Batteries: Improving Sustainability / Energy Density

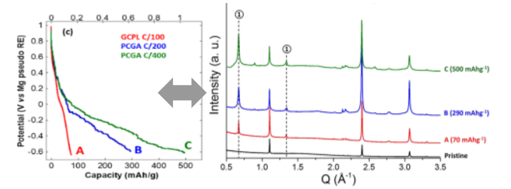
Roberta Verrelli

MATENER2018

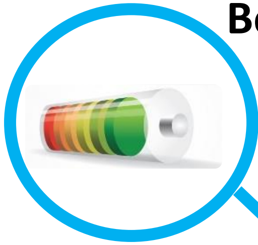
Outline

Lab-Scale

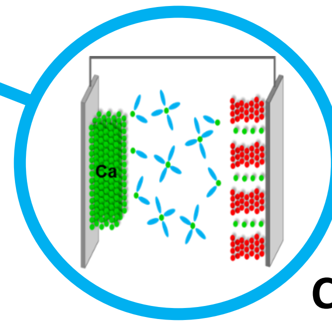
Fundamental Understanding of the Reaction Mechanisms



Beyond Li-Ion Batteries

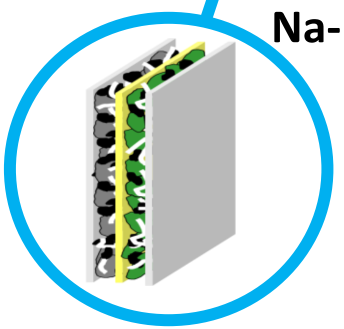


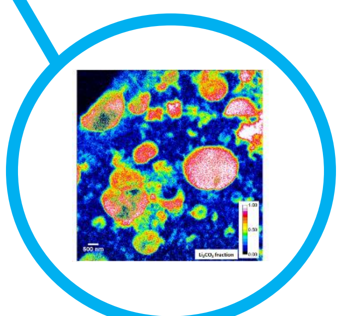
Research at ICMAB



Ca & Mg

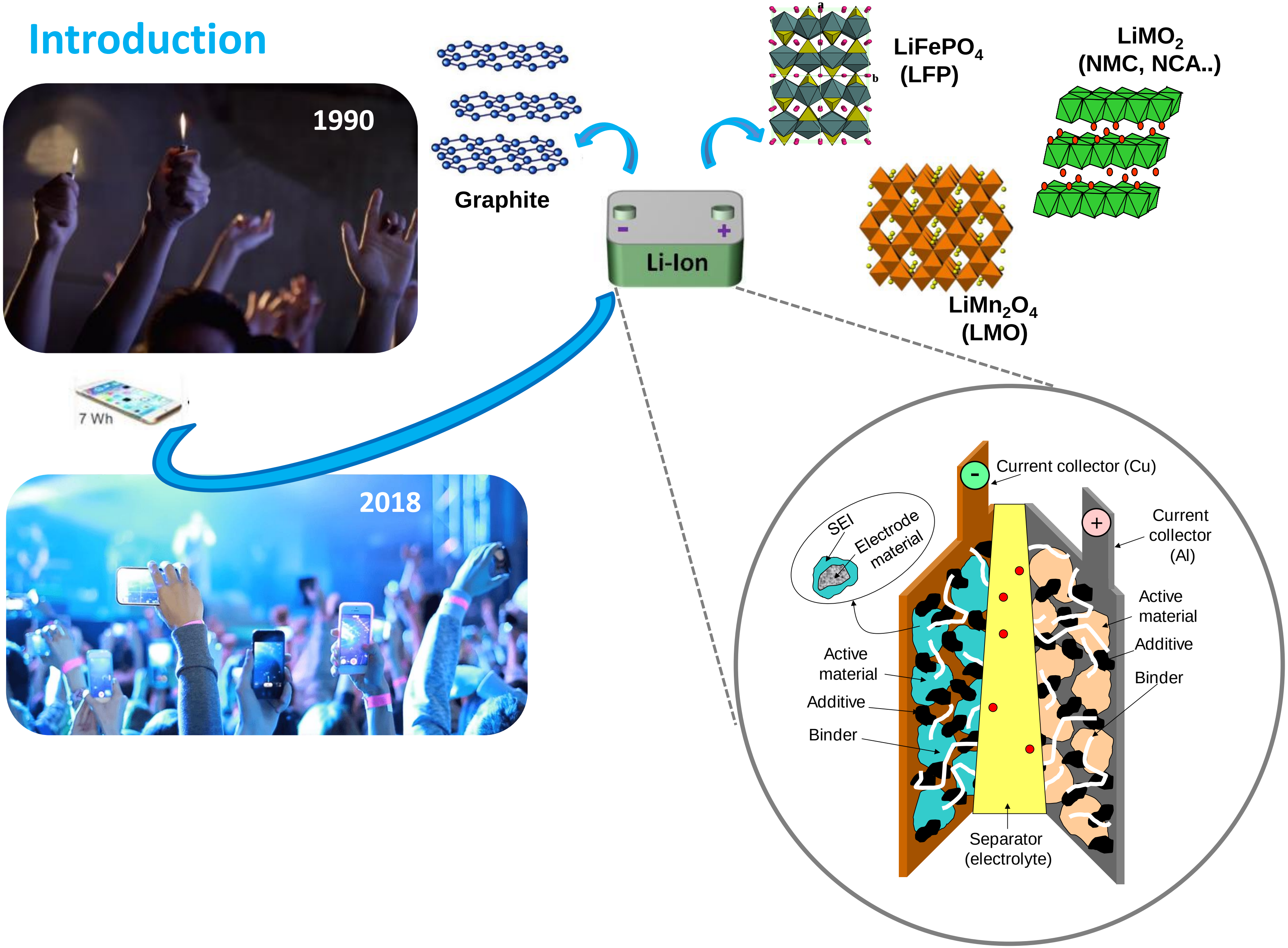
Na-Ion





M-O_2
M=Li, Na, Zn

Introduction



Larcher, D.; Tarascon, J. *Nat. Chem.* **2015**, *7*, 19–29
Croguennec, L., Palacin M. R., *J. Am. Chem. Soc.*, **2015**, 137 (9), 3140–3156.

Introduction

Performance

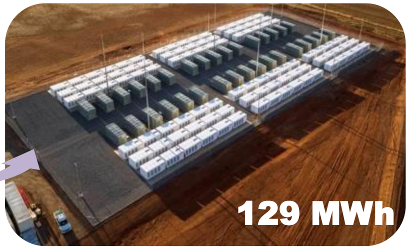
~ · 10

~ · 600

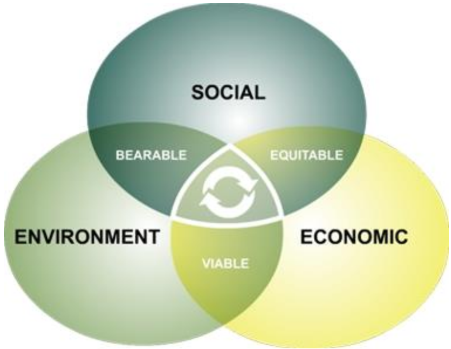
~ · 18 000 000

Application

4.10 KWh



Sustainability



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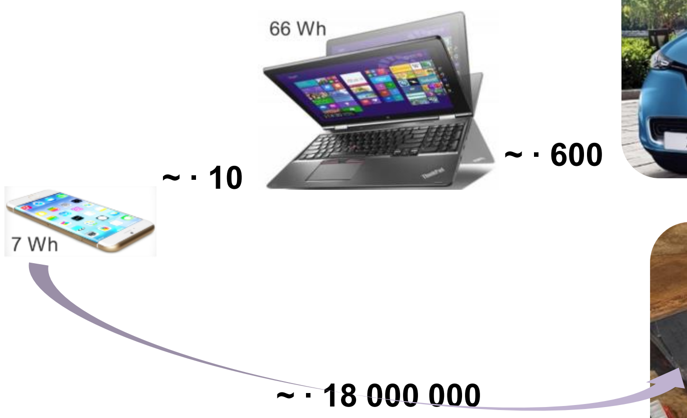
Introduction

Performance



Application

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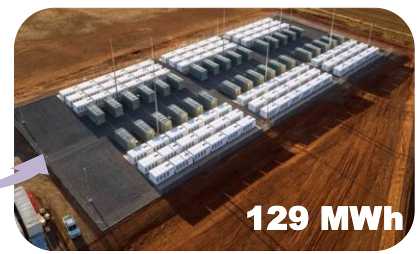
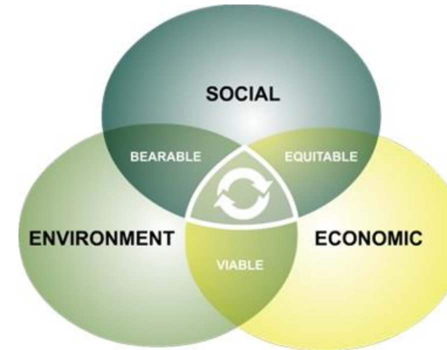


Sustainability

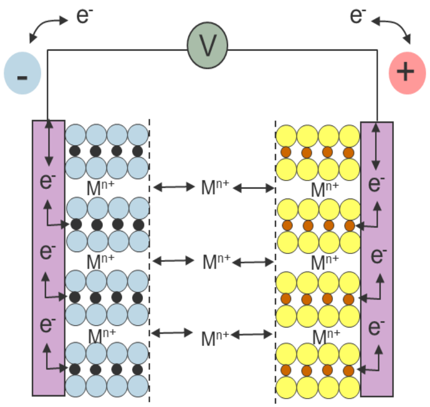


Towards more abundant elements

Larcher, D.; Tarascon, J. *Nat. Chem.* **2015**, *7*, 19–29

Beyond Li-Ion Batteries

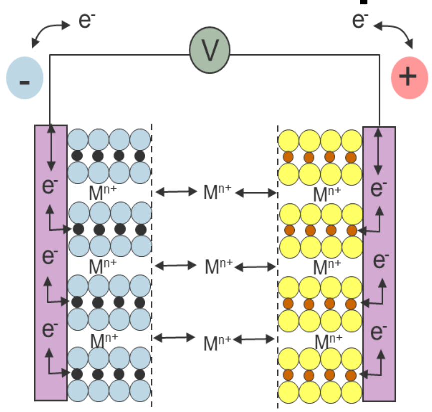
M-ion concept



Analogous to Li-ion → **Na-ion**

Beyond Li-Ion Batteries

M-ion concept

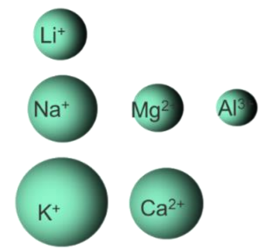


Analogous to Li-ion → Na-ion

M^{n+} as charge carriers instead of Li^+

Increased Capacity

BUT

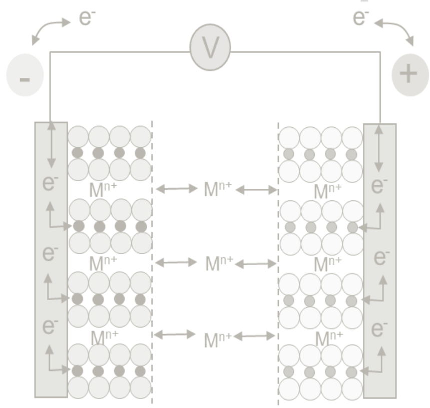


Poor Kinetics
High de-solvation Barriers

P. Canepa, G. Sai Gautam, D. C. Hannah, R. Malik, M. Liu, K. G. Gallagher, K. A. Persson, G. Ceder, *Chem. Rev.*, 2017, **117**, 4287–4341.

Beyond Li-Ion Batteries

M-ion concept



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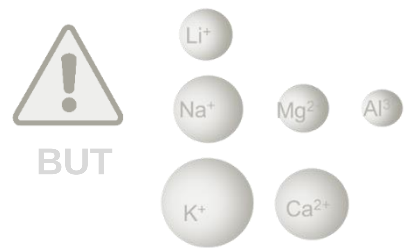
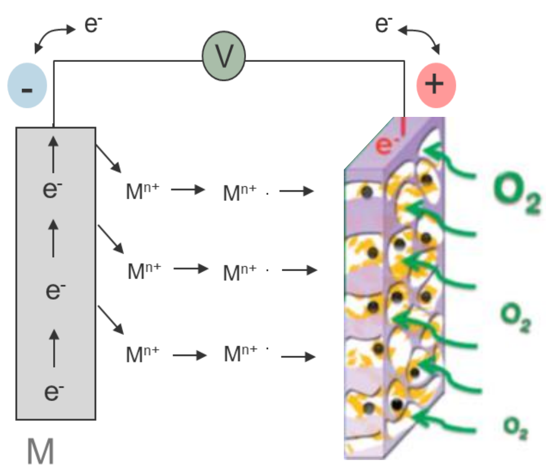
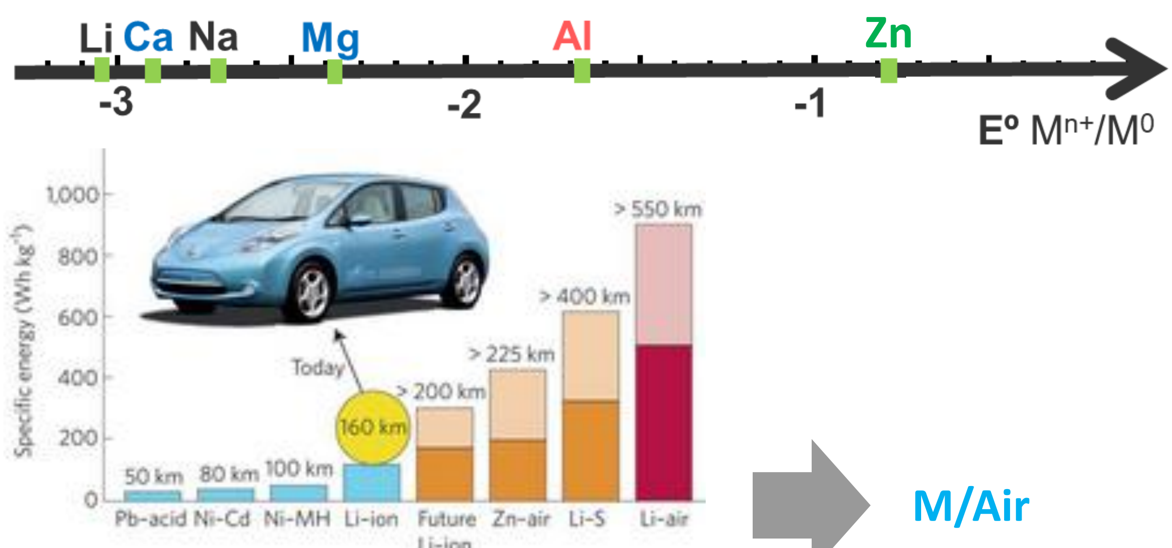


BUT

Poor Kinetics
High de-solvation Barriers

M-anode concept



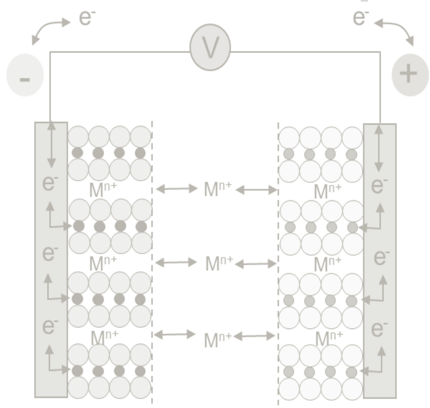


M/Air

P.G. Bruce et al. Nature Materials 11, 19–29 (2012)

Beyond Li-Ion Batteries

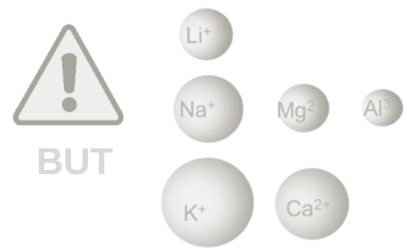
M-ion concept



Analogous to Li-ion → Na-ion

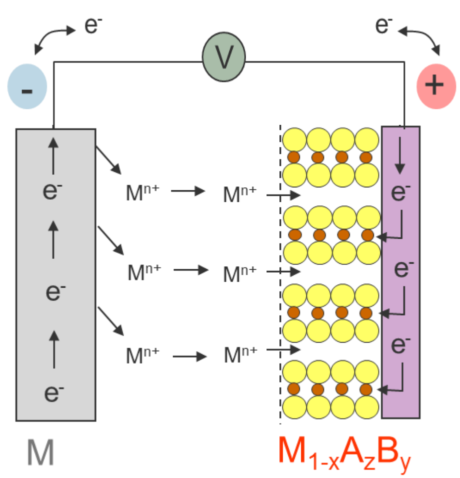
M^{n+} as charge carriers instead of Li^{+}

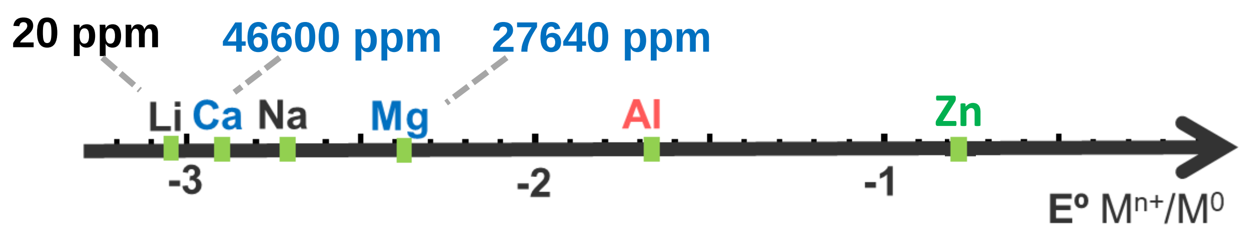
Increased Capacity

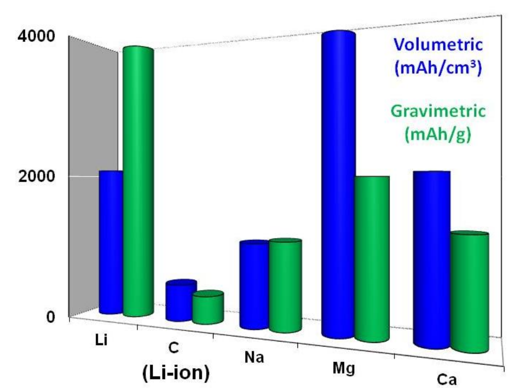


Poor Kinetics
High de-solvation Barriers

M-anode + M-Ion concept







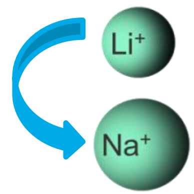
Ca & Mg

P. Canepa, G. Sai Gautam, D. C. Hannah, R. Malik, M. Liu, K. G. Gallagher, K. A. Persson, G. Ceder, *Chem. Rev.*, 2017, **117**, 4287–4341.

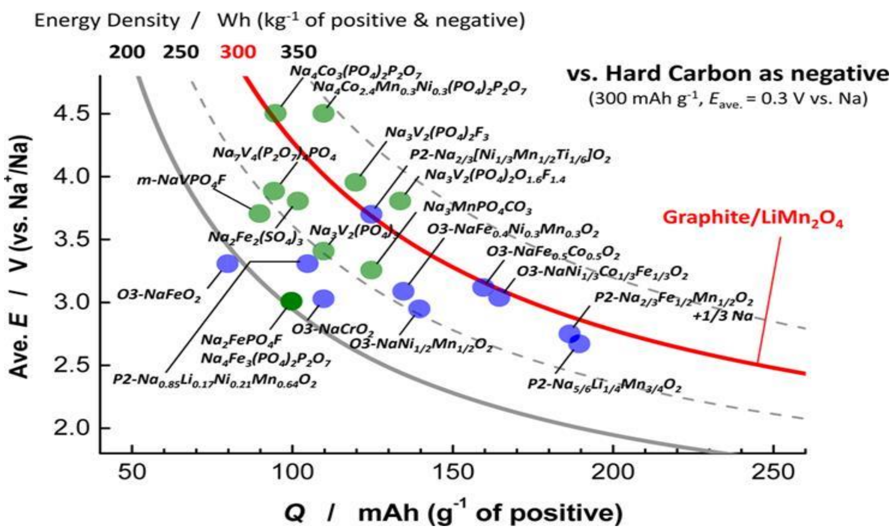
Na-Ion Batteries

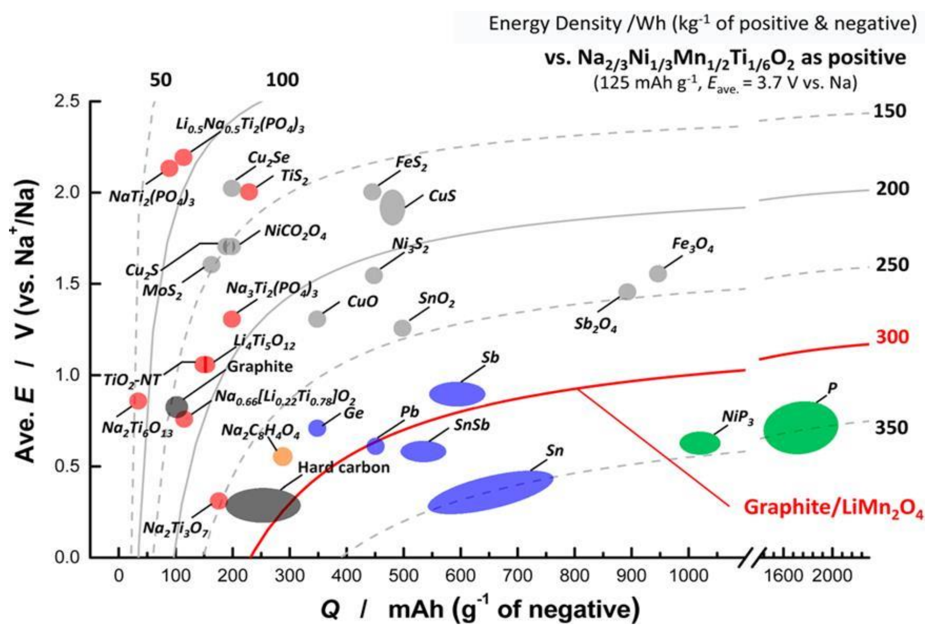
- Natural Abundance, Low Cost
- High standard reduction potential (-2.71 V vs. NHE)
- No alloy formation with Al

Directly transferable know-how



- Ionic radius
- Coordination/Crystal Chemistry
- Polarizing character, Kinetics
- Solvation energy, Solubilities, SEI



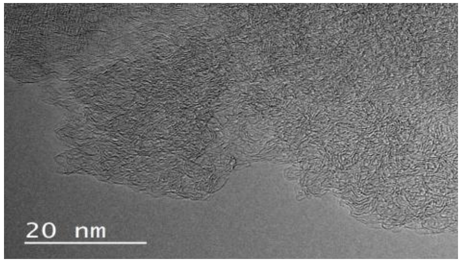


PI: M.R. Palacín

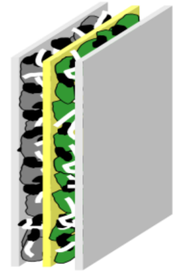
N. Yabuuchi Chem. Rev., 2014, 114 (23), 11636–11682

Na-Ion Batteries

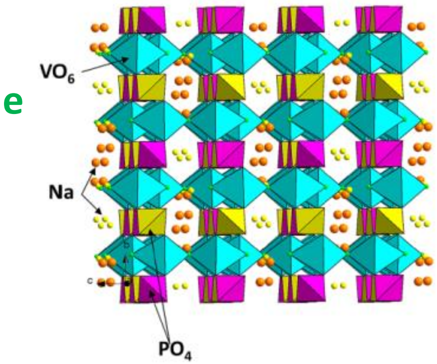
Hard C // 1 M $NaPF_6$, $EC_{0.45}$:$PC_{0.45}$:$DMC_{0.1}$ // **$Na_3V_2(PO_4)_2F_3$**



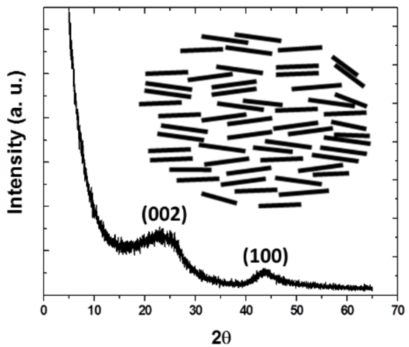
Non-graphitizable Anode
(sp^3 cross-linking)



Polyanionic Cathode



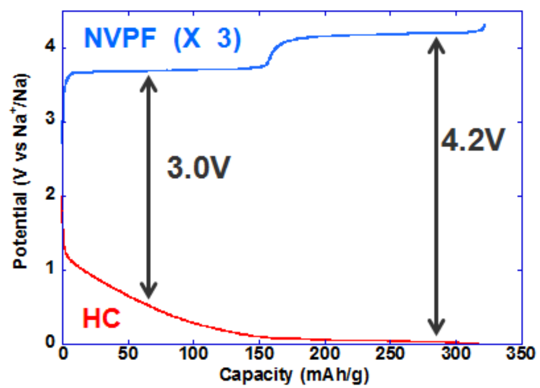
Theoretical energy density : 78 Wh/kg

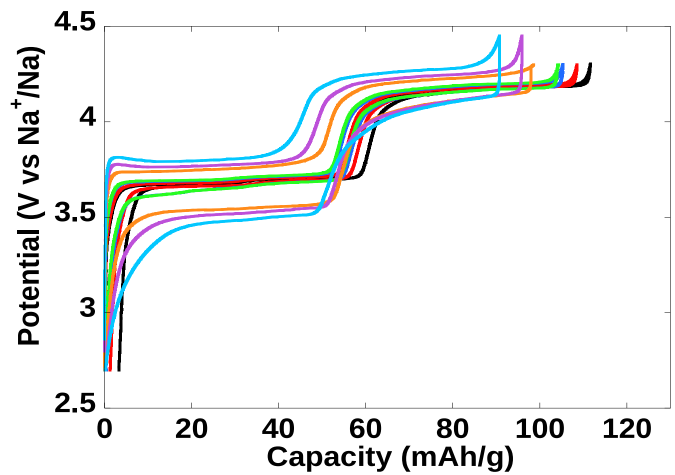


graphene sheets
10 – 40 Å

Prepared by pyrolisis
of solid precursors
(cellulose, charcoal, phenolic resins,
sugar…)

3.6 V cell
300 mAh/g (HC)
110 mAh/g (NVPF)





A. Ponrouch, R. Dedryvère, D.Monti, A. E. Demet, J. M. Ateba Mba, L. Croguennec, C. Masquelier, P. Johansson, M. R. Palacín, Energy & Environ. Sci. 6 (**2013**) 2361.

E. Irisarri, A. Ponrouch, M. R. Palacin J. Electrochem. Soc. 162 (**2015**) A2476

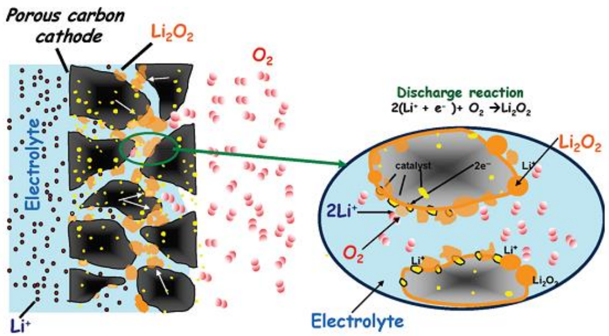
PI: M.R. Palacín

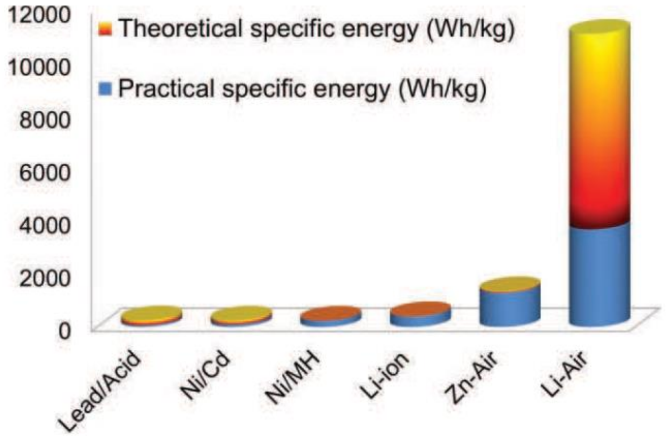
Lab Upscaling

1000 Wh

PI: M.R. Palacín

Metal-O_2 Batteries



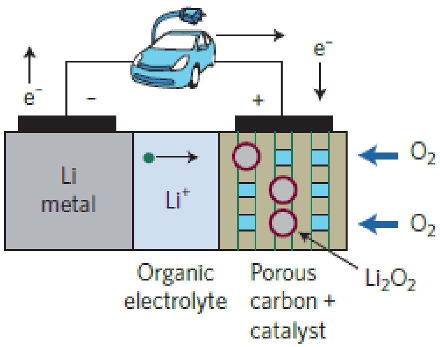


Aprotic Li-O_2

$2Li \rightarrow 2Li^+ + 2e^-$

$O_2 + 2Li^+ + 2e^- \rightarrow Li_2O_2\downarrow$

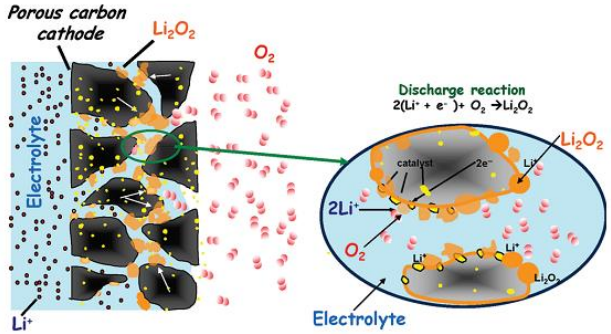
$2Li + O_2 \rightarrow Li_2O_2\downarrow$

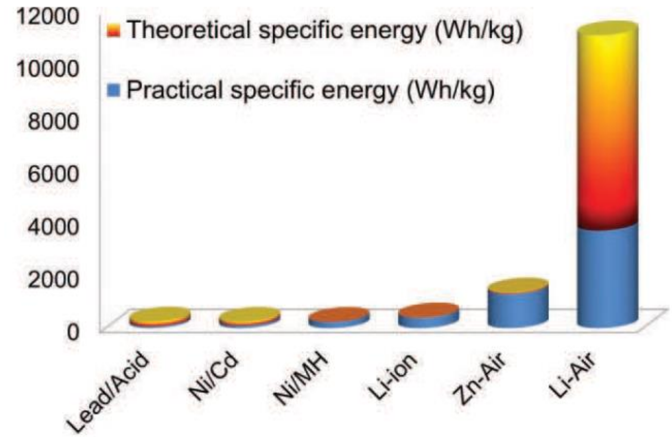


P.G. Bruce et al. Nature Materials 11, 19–29 (2012)

PI: D. Tonti

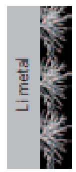
Metal-O_2 Batteries

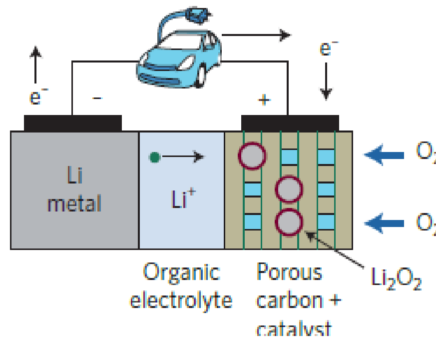


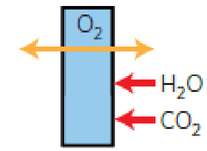


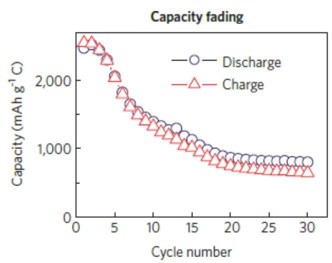
Anode

Problems of Li metal
- Dendrite formation
- Cycling efficiency
- Requires stable solid-electrolyte interphase
- Safety issues

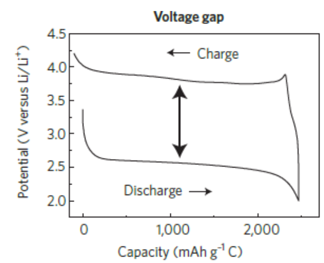








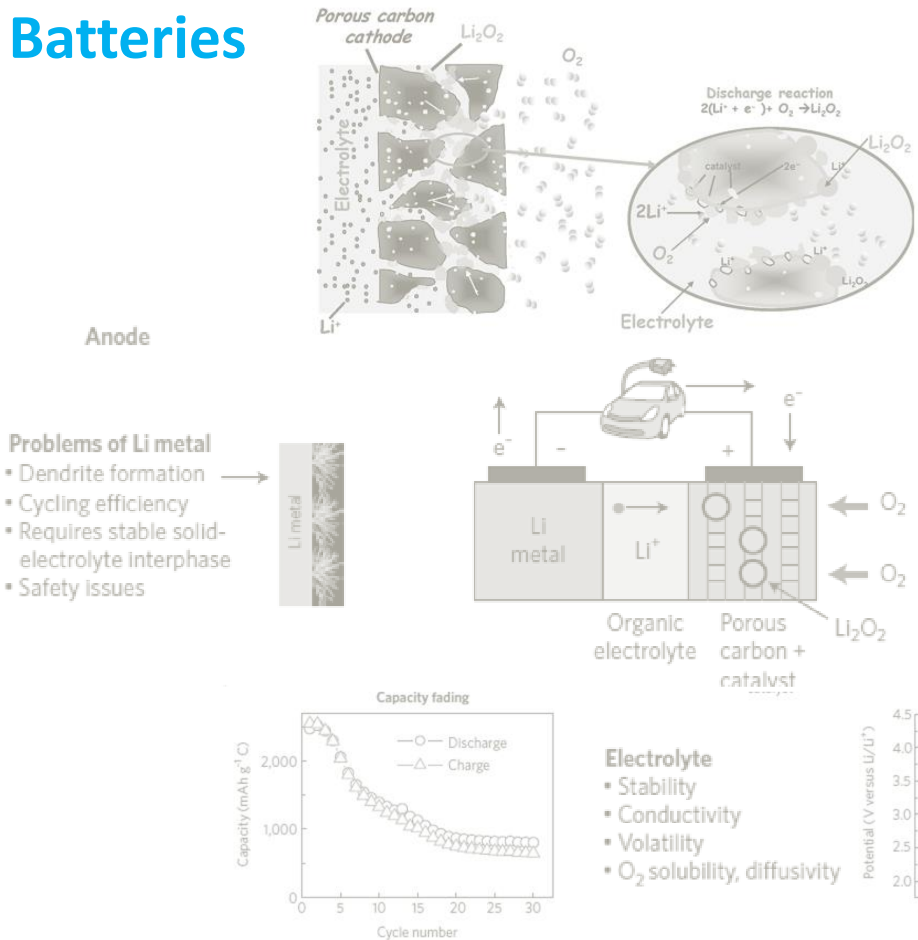
Electrolyte
- Stability
- Conductivity
- Volatility
- O_2 solubility, diffusivity

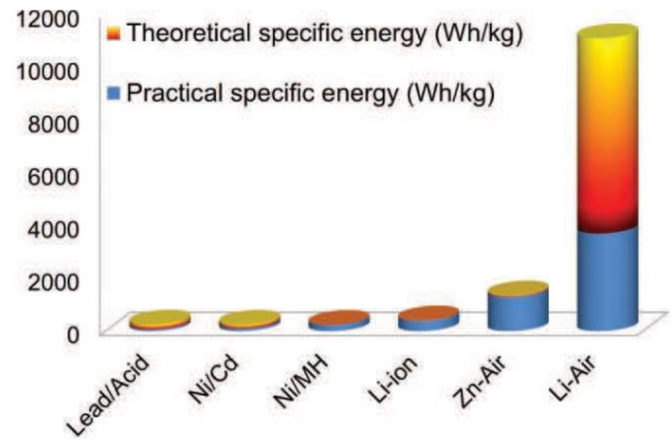


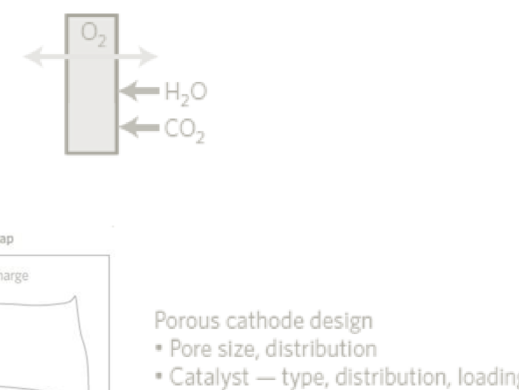
Porous cathode design
- Pore size, distribution
- Catalyst — type, distribution, loading

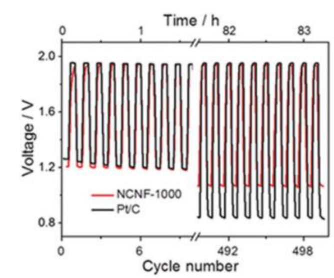
PI: D. Tonti

Metal-O_2 Batteries









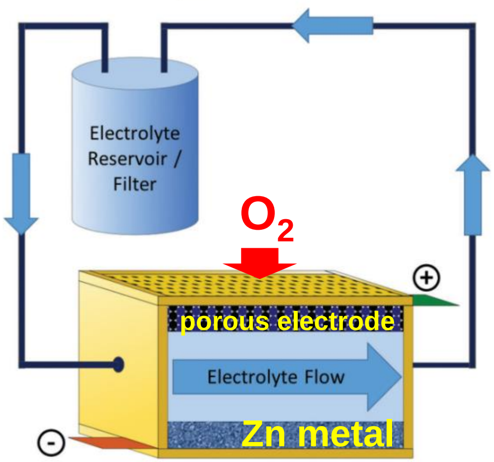
Aqueous Zn-O_2

$2Zn + 4OH^- \rightarrow 2Zn(OH)_4^{2-} + 4e^-$

$Zn(OH)_4^{2-} \rightarrow ZnO + H2O + 2OH^-$

$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

$\mathbf{2Zn + O_2 \rightarrow 2ZnO}$



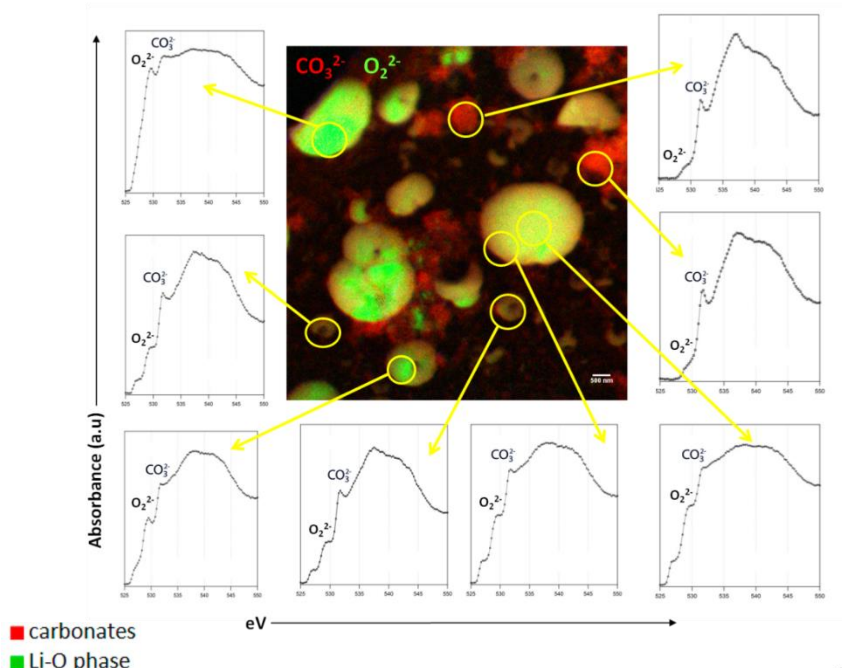
- Soluble discharge intermediate: no passivation/clogging
- Less severe dendrite formation and stability issues
- Longer cycle life

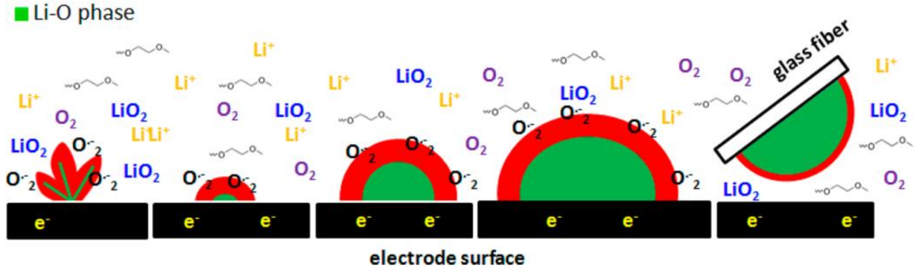
PI: D. Tonti

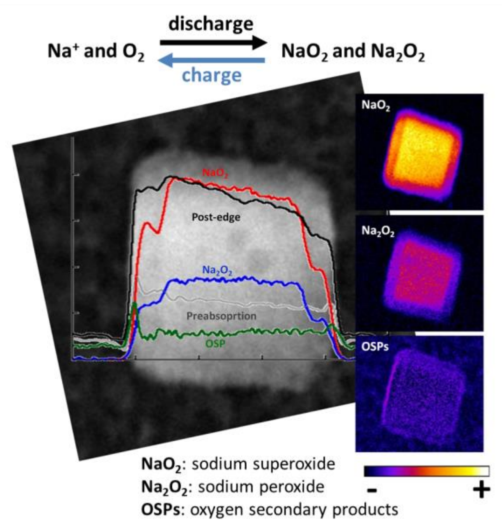
Soft X-Ray Transmission Microscopy on Li- and Na-O_2 Batteries

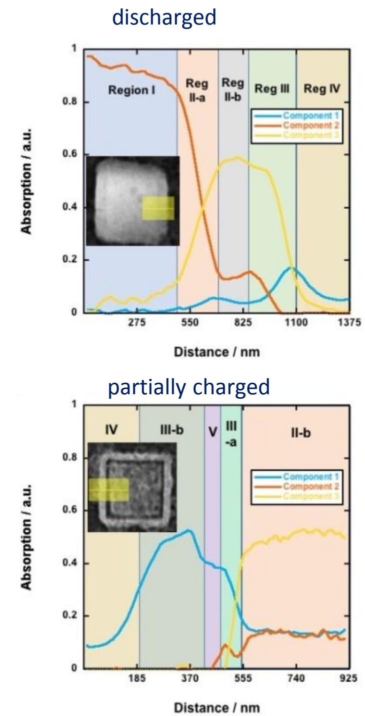
- Spectroscopic access to the oxygen chemical state
- Distribution of oxygen discharge products elucidated
- LiO_2 observed on large Li_2O_2 toroidal particles (not on smaller deposits)
- Presence of Na_2O_2 in NaO_2 cubic particles
- Carbonates irregularly distributed and covering Li_2O_2 and NaO_2 particles

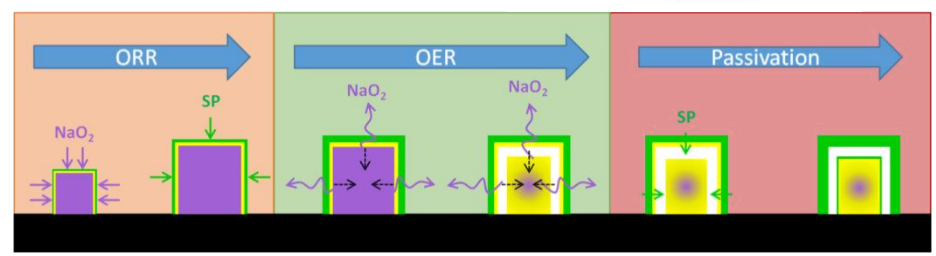
PI: D. Tonti









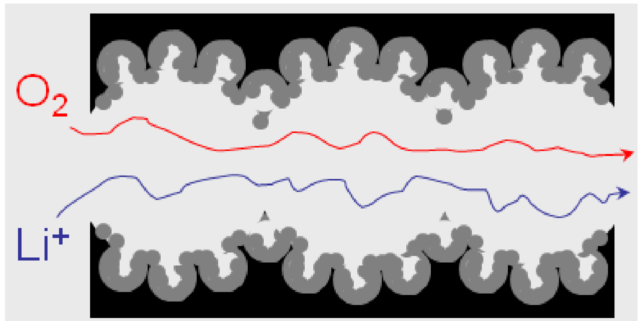


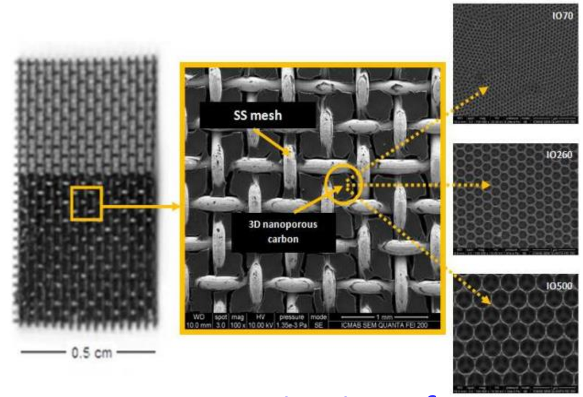
Olivares-Marín et al. *Nano Letters*, **15** (2015) 6932

I. Landa-Medrano et al. *Nano Energy* 37 (2017) 224

Li-O_2 Batteries

Cathode architectures

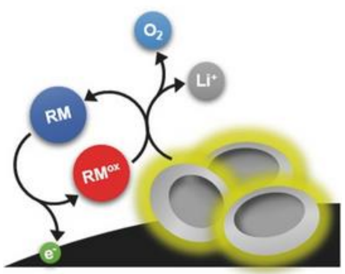


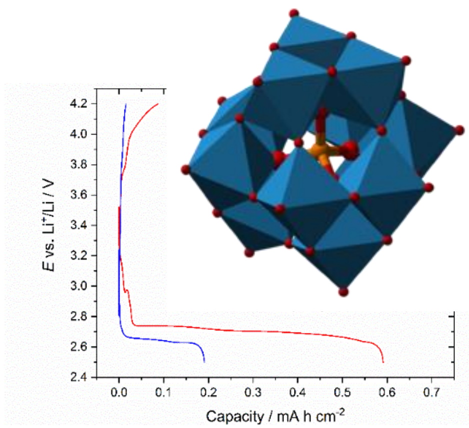


Hierarchic porosity to preserve electrolyte access to active interface

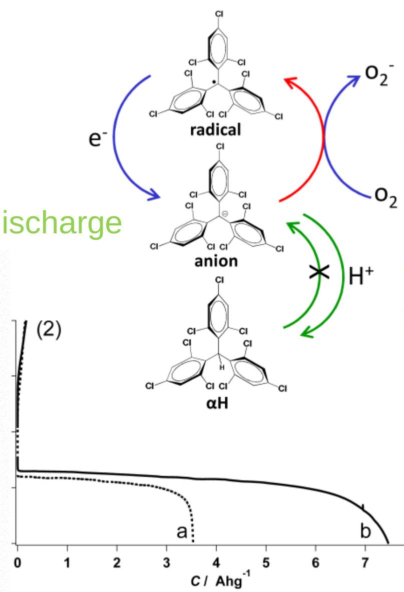
J. Mater. Chem. A 1, 14270 (2013)

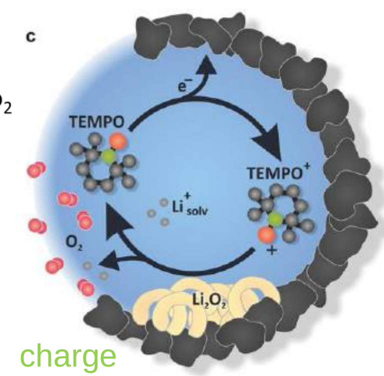
Redox Mediators





discharge





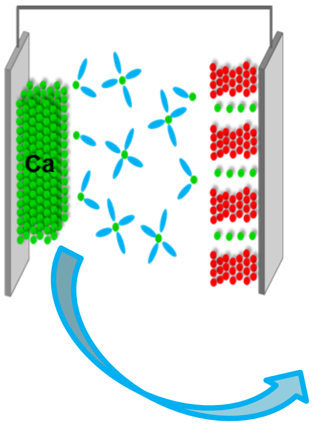
charge

Electron transport to and from the bulk: enhancing discharge and enabling full charge

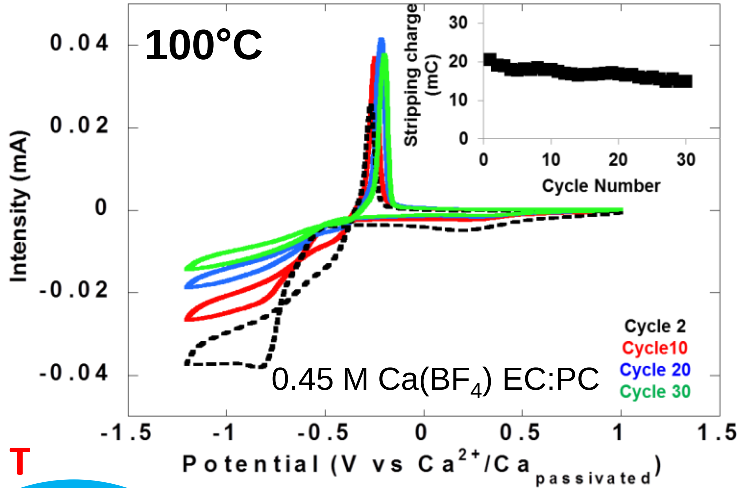
Y. Tesio et al., Chem. Commun. 51, 17623 (2015)
I. Landa-Medrano et al., Electrochem. Commun. 59, 24 (2015)
T. Homewood et al. Chem Commun 54 (2018) 9599

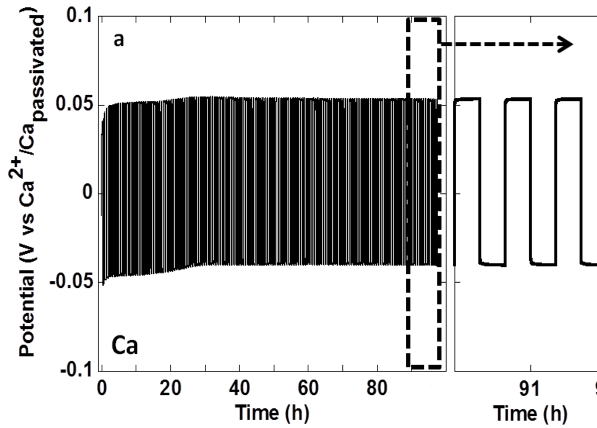
PI: D. Tonti, N. Casañ

Multivalent Batteries: Ca Metal Anode



High T



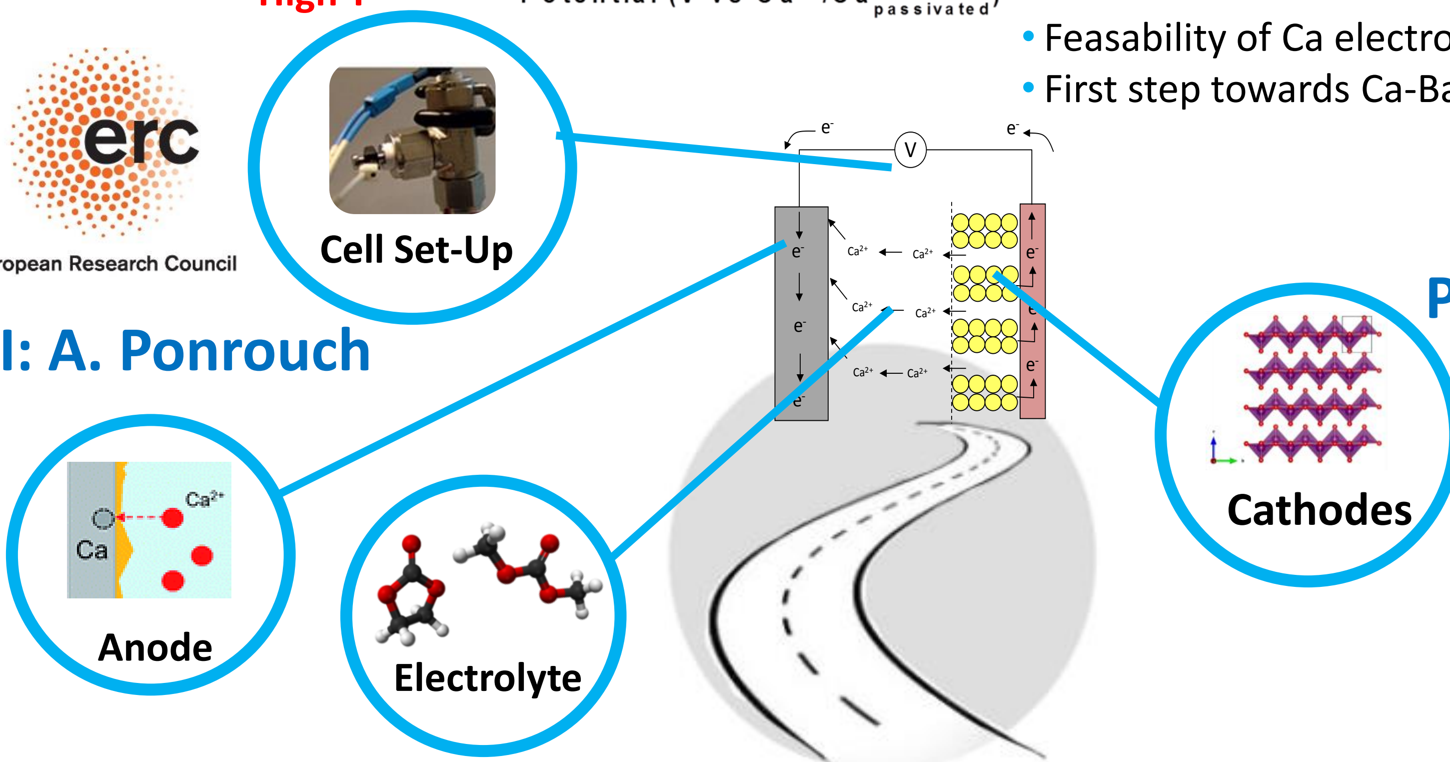


- Feasability of Ca electrodeposition
- First step towards Ca-Battery proof of concept

European Research Council

PI: A. Ponrouch

PI: M.R. Palacín



CARBAT

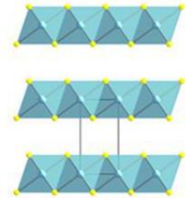
TOYOTA

HORIZON 2020

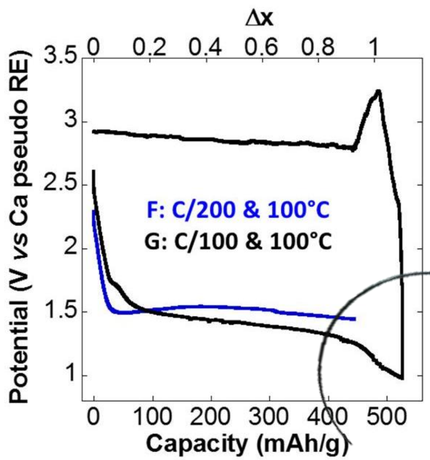
A. Ponrouch, C. Frontera, F. Barde, M.R. Palacin *Nature Materials* 15, 2016, 169

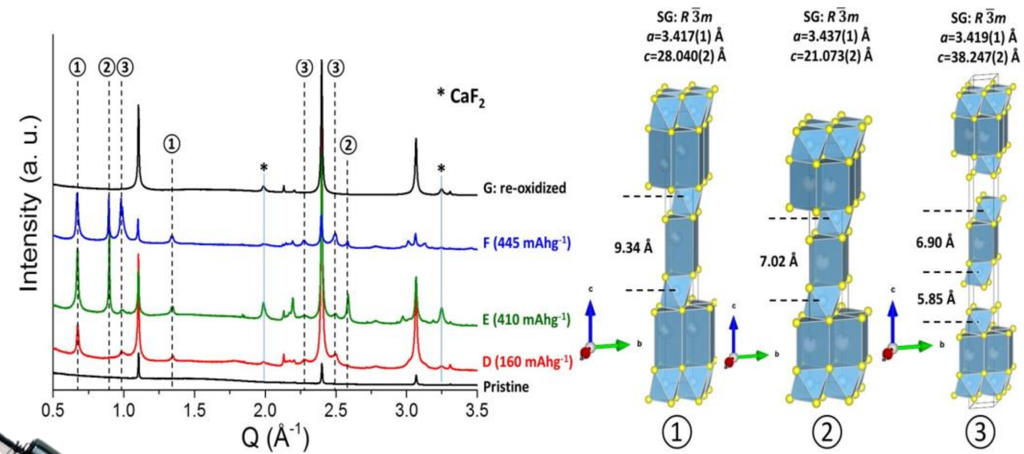
Multivalent Batteries: The Quest for Cathode Materials

Revisting Traditional Intercalation Hosts ⟹ TiS_2

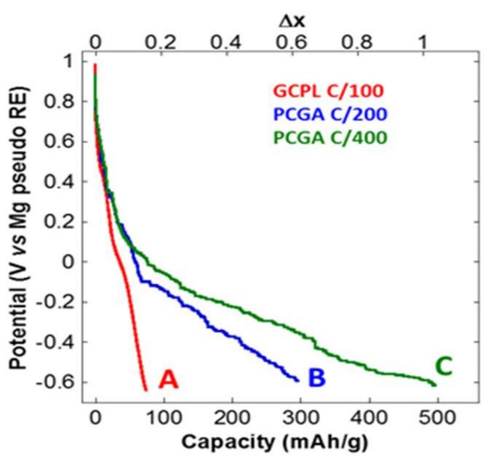


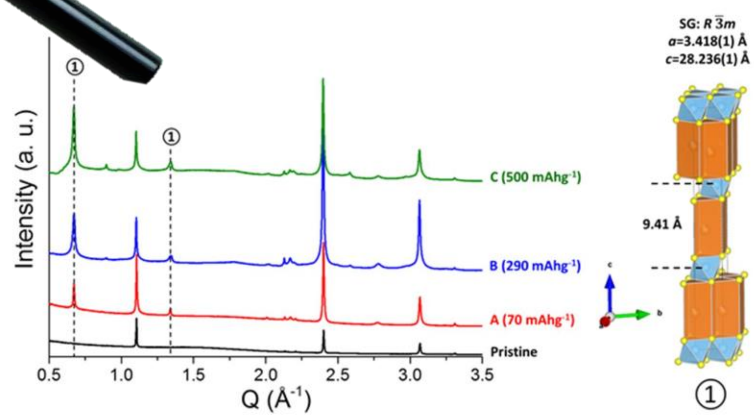
20 Ca Calcium 40.078





12 Mg Magnesium 24.305



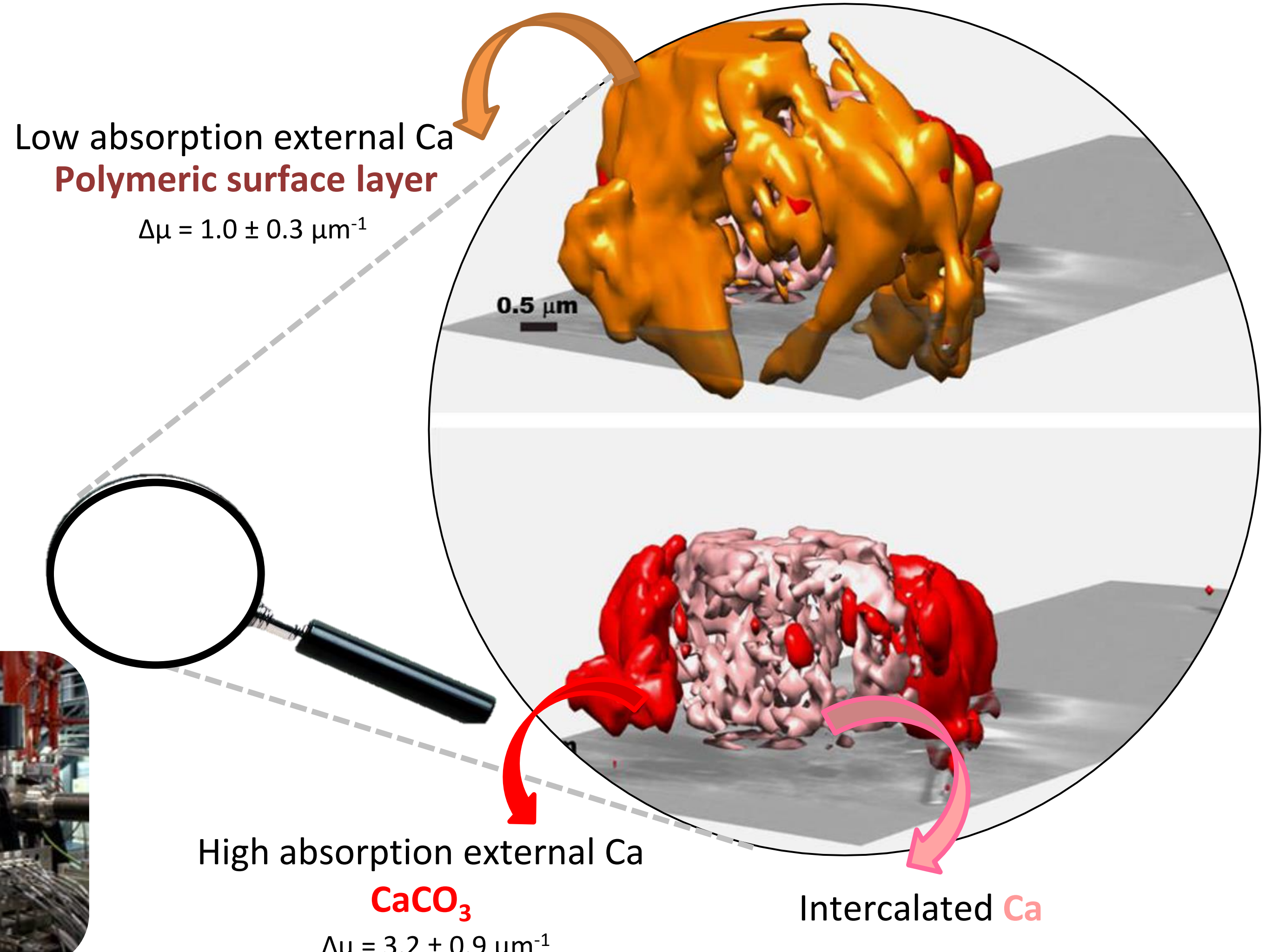


PI: A. Ponrouch & M.R. Palacín

D. Tchitchekova, A. Ponrouch, R. Verrelli, T. Broux, C. Frontera, A. Sorrentino, F. Barde, N. Biskup, M.E. Arroyo-de Dompablo, M.R. Palacín, *Chem. of Materials* ,2018, **30**, 847.

Multivalent Batteries: The Quest for Cathode Materials

Differential Absorption
X-Ray Tomography
Ca L_2 edge



Dr. A. Sorrentino

PI: A. Ponrouch & M.R. Palacín

D. Tchitchekova, A. Ponrouch, R. Verrelli, T. Broux, C. Frontera, A. Sorrentino, F. Barde, N. Biskup, M.E. Arroyo-de Dompablo, M.R. Palacín, *Chem. of Materials* ,2018, **30**, 847.

Overall Battery R&D Process: from Conception to Production

Concept Generation → **Production**

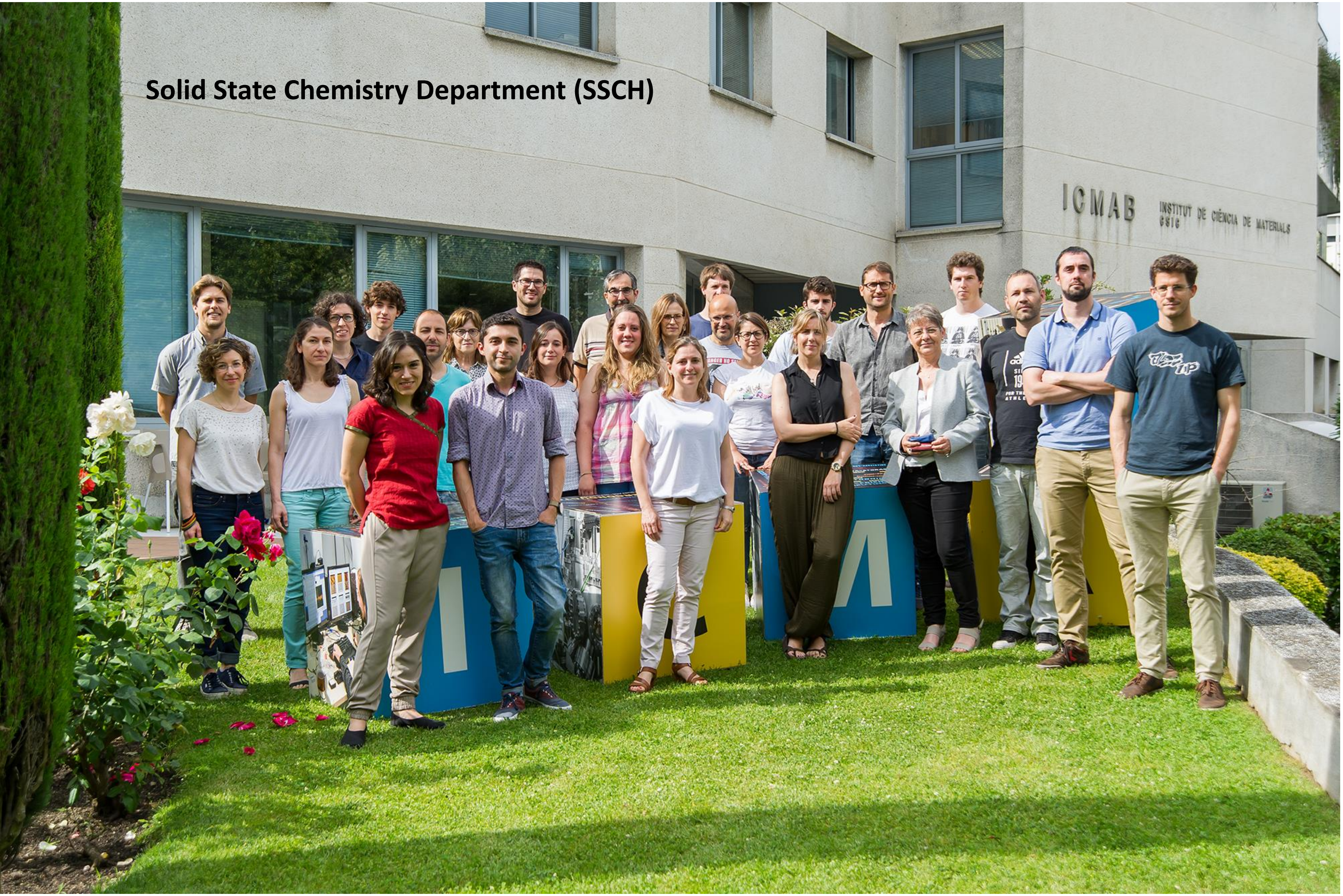
	Concept Validation	Research	Applied Research	Development	Advanced Development
	An idea in a creative mind Limited exploratory laboratory experiments Establish repeatability of performance Is there a market?	Scale-up experiments Characterize fundamental properties of concept, chem. composition, structure, etc. Evaluate size of commercial opportunity	Lab/prototype cells Initial map of performance, rate, cycling, temperature, etc. Scale-up of material preparation Preliminary market scope	Confirm research results Establish initial product format Develop unit assembly operations Make, test, and characterize 5 to 10 cell lots of 100 cells each Construct business plan	Design initial cell product Design and construct unit operations Scale-up prototype cell fabrication Run 3 to 5 sizable pilot line-factory trials Finalize business plan Market development
Timing	One to three years	One to three years	Three to four years	Three to five years	Two to four years
Staffing	One	Two to four	Four to ten	Eight to sixteen	Twelve to thirty
Materials Batch	Grams	10 to 50 g	100 g to 1 kg	1 kg to 10 kg	10 kg to 100 kg

10-19 years!!

R. Brodd, ATP Working Paper Series Working Paper 05–01, NIST (US)

Aknowledgements

Solid State Chemistry Department (SSCH)



Thanks for your kind attention